

SPACE SEGMENT (Real-time Prediction & Dynamic BFD)

Onboard

Input

- Uplink Profile
 - Model Coefficients

- Timestamp
- Coefficients
- Link Features

Coeffs
→

3. Onboard Predictor Real-time

Read uplink coefficients and predict in real time.

$$RTT_{\text{pred}} = f(t_{\text{off}})$$

where

$$f(t) = at^2 + bt + c + A \sin(\omega t + \phi)$$

Advantage:

Zero storage pressure, simple formula evaluation with minimal computing.

RTT_{pred}
Prediction
→

4. Adaptive BFD Controller Adaptive

Dynamic Tx (T_x)

Adjust transmission interval based on RTT_{pred} .

$$T_x = g(RTT_{\text{pred}})$$

(e.g., $T_x = \max(T_{\text{min}}, \alpha \cdot RTT_{\text{pred}})$)

Dynamic Timeout (T_{out})

Compute adaptive detection threshold.

$$T_{\text{out}} = (DM \times T_x) + J_{\text{max}}$$

where DM is detection multiplier, J_{max} is jitter margin.

T_{out}
(Dynamic Threshold)
→

5. Fault Detector RFC 5880

Input: Real packet stream with actual jitter & loss.



Decision: Compare actual arrival interval with dynamic threshold.

$$\Delta t_{\text{arrival}} \geq T_{\text{out}}$$

Goal:

Reduce false alarms while maximizing detection sensitivity.

State
→

BFD Session

UP / DOWN

UP

DN

Parameter Uplinking Uplink

Compressed model coefficients are sent via uplink — extremely small data volume.

< 100 bytes Per-window

UPLINK

UPLINK

GROUND SEGMENT (Offline Computation & Model Compression)

Offline

1. Orbit & Visibility Analysis

Input: Satellite Ephemeris (TLE / Ephemeris)

Compute:

- Relative position $\vec{r}(t)$
- Relative velocity $\vec{v}(t)$
- Propagation delay:

$$RTT_{\text{true}}(t) = \frac{2d(t)}{c}$$

Process: Extract continuous Visibility Windows.

True RTT Trajectory
(High-rate sample)
→

2. Modeling & Validation

Core Algorithm:

Harmonic / Polynomial Regression

$$f(t) = at^2 + bt + c + A \sin(\omega t + \phi)$$

Quality Control:

$$\text{Max Error} = \max_i |RTT_{\text{true}}(t_i) - f(t_i)|$$

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (RTT_{\text{true},i} - f_i)^2}$$

● RMSE < 0.5 ms ● Max Err < 2.0 ms

Ensure model accuracy is sufficient to support threshold decision.

Profile
→

Output:
Window-level Uplink Profile

- Timestamp
- Model Coefficients
- Link Features

